Math 126

Extra Unit A practice

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity A and Jubilee 1.

Derivatives

$$\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\left(\cos^{-1} x\right) = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1+x^2}$$
$$\frac{d}{dx}\left(\sinh^{-1} x\right) = \frac{1}{\sqrt{1+x^2}}$$
$$\frac{d}{dx}\left(\cosh^{-1} x\right) = \frac{1}{\sqrt{x^2-1}}$$
$$\frac{d}{dx}\left(\tanh^{-1} x\right) = \frac{1}{1-x^2}$$

Integrals

$$\int u\,dv = uv - \int v\,du$$
$$\int \tan x\,dx = \ln|\sec x| + C$$
$$\int \sec x\,dx = \ln|\sec x + \tan x| + C$$
$$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$$

Hyperbolic functions

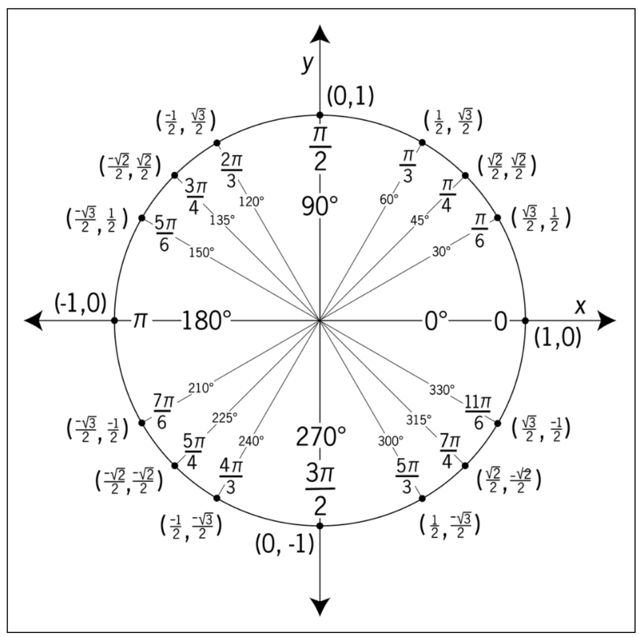
$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$
$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$
$$\sinh^{-1} x = \ln(x + \sqrt{x^2+1})$$
$$\cosh^{-1} x = \ln(x + \sqrt{x^2-1})$$
$$\tanh^{-1} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

Identities

$$\cos^2 x + \sin^2 x = 1$$
$$\tan^2 x + 1 = \sec^2 x$$
$$\sin^2 x = \frac{1}{2}(1-\cos(2x))$$
$$\cos^2 x = \frac{1}{2}(1+\cos(2x))$$
$$\cosh^2 x - \sinh^2 x = 1$$
$$1 - \tanh^2 x = \text{sech}^2 x$$

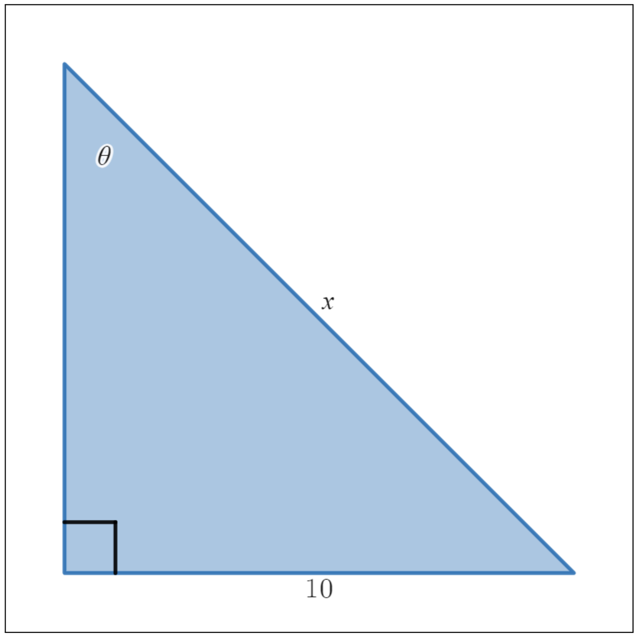
Approx. integration

$$|E_T| \le \frac{K(b-a)^3}{12n^2}$$
$$|E_M| \le \frac{K(b-a)^3}{24n^2}$$
$$|E_S| \le \frac{L(b-a)^5}{180n^4}$$



A1 - Inverse trigonometric and hyperbolic functions

(a) (i) Find a function $\theta(x)$ expressing the acute angle θ in the right triangle to the right in terms of the length of the hypotenuse, x.



(ii) Find $\theta'(x)$. If you aren't able to answer part (i), you can still receive credit for this question by finding the derivative of $\cos^{-1}(\tan(x))$.

(b) Show each of the following identities using the definitions of hyperbolic functions in terms of e^x and e^{-x}.

(i) $\dfrac{d}{dx}(\cosh(x)) = \sinh(x)$

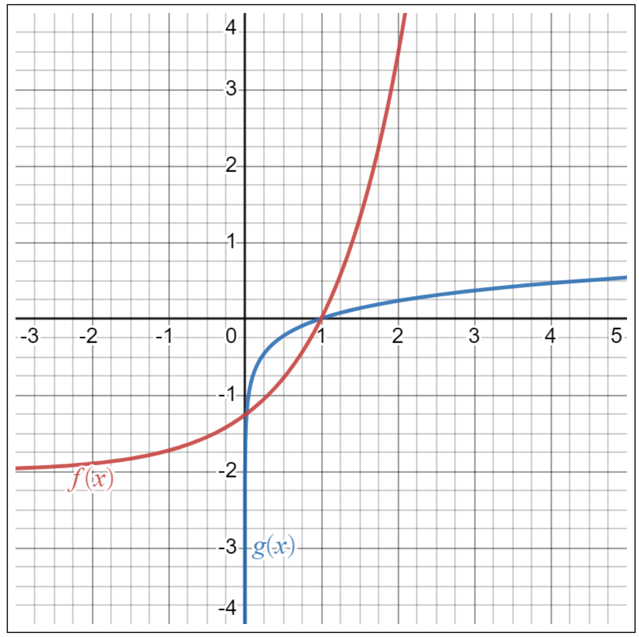
(ii) $\cosh^2(x) - \sinh^2(x) = 1$

A2 - L'Hospital's rule

(a) Two functions $f(x)$ and $g(x)$ are graphed to the right. Use the graph to estimate

$$\lim_{x \to 1} \frac{f(x)}{g(x)}$$

Indicate very clearly how you determined your estimate.



(b) Evaluate $\lim\limits_{x \to 1^+} x^{\frac{1}{1-x}}$

A3 - Integration techniques I

Evaluate each of the following integrals.

(a) $\displaystyle\int e^{2x}\cos(3x)dx$

(b) $\displaystyle\int_0^1 \frac{dx}{(x^2+1)^2}$

A4 - Integration techniques II

(a) Evaluate the integral $\displaystyle\int \frac{2x^3 + 4x - 4}{x(x^2+2)^2}\,dx$

(b) Find the values of p such that $\displaystyle\int_0^{\infty} \frac{dx}{x^p}$ converges. If no such p exists, show why.

A5 - Approximate integration

Consider the integral $I = \int_0^8 e^{-\frac{x}{2}}\,dx$. You do not need to simplify your answer for any question below.

(a) Use the Trapezoidal Rule approximation with $\Delta x = 2$ to approximate I.

(b) How many trapezoids should you use in a Trapezoidal Rule approximation to guarantee that the error will be less than 0.00001?

(c) Use Simpson's rule approximation with $\Delta x = 1$ to approximate I.

(d) Find an upper bound on the error for the approximation in part (c).